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# Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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## Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its  
2 refutation by overlapping authors; this paper joins them and locates the edge  
3 of the set that refutation relies on. Conformal PID states long-run coverage for  
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the  
5 deployed threshold inside an admissible radius. The claim was acted on; placement  
6 and derivation are conceded by name. At the null scorecaster that condition is *tight*.  
7 A legal  $L_1$  dead band on the *completed* threshold leaves that set at its radius, and  
8 past it miscoverage is 1.000000 against  $\alpha = 0.1$ . Coverage goes, not only the rate.  
9 The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the  
10 condition is sufficient only. Under the equally legal  $\hat{q} \equiv +b/2$  the failing band  
11 covers; at  $\hat{q} \equiv -b/2$  both it and partial adjustment at  $w = 0.999$  miscover.

## 1 Introduction

13 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,  
14 by the same first and last authors, proves in the same notation that a bounded perturbation of the  
15 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the  
16 joining is what this paper adds.

17 An online conformal method [Angelopoulos and Bates, 2021] emits a threshold before each obser-  
18 vation and moves it. Forecast stability names, scores and prices that movement [Tunc et al., 2013,  
19 Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026,  
20 Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification,  
21 matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012,  
22 Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al.,  
23 2026, Vaze, 2026]. None of it is claimed here.

24 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-  
25 gelopoulos et al., 2023] adds a *scorecaster*  $\hat{q}_{t+1}$  to a saturating error integrator, asking only that it be  
26 predictable and lie in  $[-b/2, b/2]$ : “any function of the past” (Theorem 1). Such a readout is already  
27 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical  
28 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-  
29 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]  
30 add any predictable  $d_t$ ,  $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$  with  $\mu_t \in [0, 1]$  predictable, to the deployed threshold;  
31 since  $\hat{q}_t + d_t$  still lies in  $[-b/2, b/2]$  it is itself a legal scorecaster, so the result “has exactly the CPID  
32 error-integration form” and  $|E_T| = o(T)$ . The same generic-slot argument carries AcMCP’s long-run  
33 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv  
34 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:  
35 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”  
36 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**  
37 **claim neither the placement nor the derivation.**

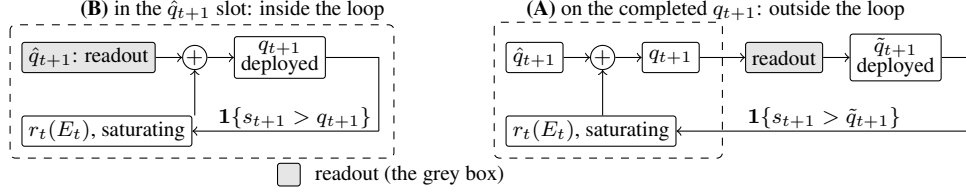


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop  $r_t$  closes. In (B) the loop closes on the integrator’s own output, and both readouts keep  $\{\hat{q}_t\}$  in the convex hull of  $\hat{q}_1$  and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on  $\tilde{q}_{t+1}$ , which it never computed, and the deployed sequence leaves that set.

38 **Where that refutation stops, the condition is tight.** Corollary 2 is a *retention* result, silent outside  
 39 its radius, and the question it leaves is whether that radius is sharp. Section 3 exhibits a legal  
 40 perturbation leaving it: the  $L_1$  dead-band family characterised here on the *completed* threshold,  
 41 whose edge is the corollary’s radius at the null scorecaster. Past it coverage goes, not only the  
 42 rate: the published sufficient condition is necessary there. Appendix C sets both claims beside  
 43 four vocabularies for the same movement; Appendix D reads  $E_t$  as a capital process and builds a  
 44 calibration test on it.

## 45 2 Setup

46 **The producer.** A forecaster fit before  $t$  induces a score  $s_t$ , and a threshold  $q_t$  is emitted before  
 47  $y_t$  is revealed. The deployed set is  $C_t = \{y : s_t(x_t, y) \leq q_t\}$  and  $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$ . The  
 48 target is long-run coverage,  $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$ , with no model on the data. Conformal PID  
 49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale, generalizing the running-  
 50 error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift  
 51 variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left( \sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

52 Angelopoulos et al.’s iteration (5). Its  $r_t$  saturates: condition (4) requires  $|r_t(x)| \geq b$  with the sign of  
 53  $x$  once  $|x| \geq c h(t)$ , for  $h$  nonnegative, nondecreasing and sublinear. With  $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$ ,  
 54 Proposition 2 gives  $|E_T| \leq c h(T) + 1$ , read on realised scores.

55 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard  
 56 readouts, both prior work. A quadratic cost gives linear partial adjustment  $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$   
 57 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied  
 58 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,  
 59 moving only when the target is more than  $\tau$  away, then pulling back exactly  $\tau$  [Constantinides,  
 60 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A) is  
 61 measured here. *The dead band ( $L_1$ ) is not a stylistic alternative to the partial-adjustment ( $L_2$ ) form;*  
 62 *it is the family whose failure Section 3 characterises. Neither is safe by construction:  $L_2$  forfeits*  
 63 *Proposition 2’s rate while covering,  $L_1$  past the radius loses coverage, and at the legal  $\hat{q} \equiv -b/2$  both*  
 64 *lose it. What is tight is the radius, not a form.*

65 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*,  $\mathcal{T}_H =$   
 66  $\sum_{t=2}^H |q_t - q_{t-1}|$ , after actuator travel in process control. The arithmetic is not new. Over a predictive  
 67 distribution  $\sum |\Delta q|$  is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it  
 68 *jumpiness*, a *convergence index*, (in)consistency [Zsótér et al., 2009, Ehret, 2010, Pappenberger  
 69 et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length.  
 70 Table 4’s caption names ten, so a fresh one is used here.

71 **Harness.** The harness fixes  $\alpha = 0.1$  and  $b = 2$ . Its saturator  $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$ ,  
 72 with  $c = 1$  and  $h(t) = \log(t + 2)$ , meets (4) with equality. The scorecaster is null and legal, reducing  
 73 (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower  
 74 bound met exactly here, and Theorem 1 permits scorecasters  $b/2$  either side of null, predictable  
 75 but not constant in  $t$ . The adversary is specified, not tie-broken. It plays  $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ ,  
 76  $\varepsilon = 10^{-9}$ , against the *deployed*  $\tilde{q}_t$ : the  $\varepsilon \downarrow 0$  reading of “sets the score at the threshold”, firing the

77 strict indicator. Legality is Theorem 1’s  $[-b/2, b/2]$  on both sides, so  $b/2 + b/2 = b$  is exactly tight.  
 78 *That specification names one member of a class:* any adapted  $s_t \in [-b/2, b/2]$  is legal, the constant  
 79  $s_t \equiv -b/2$  included. Every number in Table 2 (App. A) is measured against the one adversary above;  
 80 Section 3’s two boundaries are quantified over the class, and they differ.

### 81 3 The edge of the admissible set, located and proved

82 **The retention result is published, and tested from outside.** Li et al. [2025, Corollary 2] prove  
 83 that a predictable  $d_t$  with  $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$  on the deployed threshold keeps (1) in error-integration  
 84 form and *retains*  $|E_T| \leq ch(T) + 1$ : the radius is theirs. A downstream partial adjustment of weight  
 85  $w$  leaves it; Table 2 (App. A) prices that departure, and five of its eleven arms exceed the bound at  
 86  $T = 10^6$ , and for  $w = 0.99$  and  $0.999$  the excursion is *flat in  $T$*  while the bound grows like  $\log T$ .  
 87 The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel  
 88 falls from 28,311 to 202. A readout on the completed  $q_{t+1}$  touches neither  $r_t$  nor the integrator’s  
 89 argument, so it cannot put (4) at risk. It delays a correction already computed, and the accumulator  
 90 excurses *further*. Smoothing the output does not damp  $E_t$ ; it feeds it.

Table 1: Placement A, primary regime,  $T = 10^6$ : four arms of Table 2 (App. A), which lists the remaining seven and finer detail. Proposition 2’s bound is 14.8155; “ratio” is  $\max_t |E_t|$  over it.

| Arm                      | miscov. $10^6$  | $\max_t  E_t  10^6$ | ratio  |
|--------------------------|-----------------|---------------------|--------|
| none (control)           | 0.100007        | 7.80                | 0.53   |
| partial adj. $w = 0.999$ | <b>0.100004</b> | 623.70              | 42.10  |
| dead band $\tau = 0.9$   | 0.100012        | 13.20               | 0.89   |
| dead band $\tau = 1.5$   | <b>1.000000</b> | 900,000             | 60,747 |

91 **The  $L_1$  dead band leaves the radius, and the edge is a theorem.** A dead band pulls the deployed  
 92 threshold back by exactly  $\tau$  when it releases and holds it otherwise, so  $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$  at every  
 93 round and “inside Corollary 2” reads  $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$ . Under saturation the deployed value  
 94 is pinned  $\tau$  below the integrator’s ceiling, and its realised supremum is  $b - \tau$  to the printed digit  
 95 at every  $\tau \geq 0.9$ . Outside the radius two levels decide the answer and condition (4) names only  
 96 one of them. Write  $A_t^+ = \sup_x r_t(x)$  and  $A_t^- = \inf_x r_t(x)$  for the integrator’s extremes, and  
 97  $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$ ,  $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$  for the levels it is *guaranteed* to reach once the  
 98 accumulator clears the radius. Condition (4) says exactly  $\Lambda_t^+ \geq b$  and  $\Lambda_t^- \leq -b$ ; that  $\Lambda_t^\pm = A_t^\pm$ ,  
 99 that the integrator reaches its extremes where (4) puts them, is a further hypothesis and not one (4)  
 100 supplies. Set

$$\tau^{\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

101 **Failure, over the whole admissible class.** If  $\tau > \tau^{++}$  then  $\hat{q}_{t+1} + A_t^+ - \tau < b/2$  at every round,  
 102 so  $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$  makes  $\{\tilde{q} < b/2\}$  absorbing: a release upward lands below  $b/2$ , and a  
 103 hold or a release downward cannot lift the deployed value at all. Against  $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$  the  
 104 indicator is then  $1\{\tilde{q}_t < b/2\}$  and every round misses. If  $\tau \geq \tau^{*-}$  the constant legal play  $s_t \equiv -b/2$   
 105 makes  $\{\tilde{q} \geq -b/2\}$  absorbing by the mirror inequality, so no round can miss and the realised rate  
 106 is 0 rather than  $\alpha$ . Either way long-run coverage is lost, with  $|E_T|$  linear in  $T$ . Nothing here asks  $\hat{q}$   
 107 to be constant, asks the extremes to be attained anywhere, or asks  $A_t^\pm$  to be finite beyond what the  
 108 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If  $\tau \leq \sigma^+$   
 109 then  $E_{t-1} \geq ch(t-1)$  forces  $q_t \geq b/2 + \tau$  and so  $\tilde{q}_t \geq b/2$ , whence  $\text{err}_t = 0$  against *every* legal  
 110 adversary; if  $\tau < \sigma^-$  the mirror forces  $\text{err}_t = 1$  below  $-ch(t-1)$ . The accumulator cannot then  
 111 leave  $[-ch(t), ch(t)]$  by more than one round, and  $|E_T| \leq ch(T) + 1$ : Proposition 2’s own bound,  
 112 retained verbatim rather than weakened to  $o(T)$ . Using condition (4) alone this already certifies  
 113  $\tau \leq b/2 + \inf_t \hat{q}_t$  and  $\tau < b/2 - \sup_t \hat{q}_t$ , which for a constant scorecaster is Corollary 2’s radius at  
 114  $\mu_t = 1$ .

115 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If  $\hat{q}_t \equiv \hat{q}$ , if  
 116  $\Lambda_t^\pm = A_t^\pm$ , and if neither depends on  $t$ , then  $\sigma^\pm = \tau^{\pm}$  and **coverage is retained against every legal**  
 117 **adapted adversary if and only if  $\tau \leq \tau^{++}$  and  $\tau < \tau^{*-}$ .** For a symmetric saturator that is one line:  
 118 coverage if and only if  $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$ , with the endpoint included only when  $\hat{q} < 0$ .  
 119 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**

120 **before is the failure half.**  $\tau^{*+}$  is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs  
 121 agree with it read against the adversary the harness plays. It is not a retention condition, and Table 3  
 122 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never  
 123 go:  $|E_t| \leq (1 - \alpha)t$ , so a saturator equal to the harness's inside  $|x| \leq t^2$  and to  $\pm 10b$  outside it  
 124 satisfies (4), reads  $\tau^{*+} = 19$ , and fails at  $\tau = 1.001$ . A legal time-varying  $\hat{q}$  splits  $\sup_t$  from  $\inf_t$ :  
 125  $\hat{q}_t = b/2$  on  $t$  a power of two and 0 elsewhere reads  $\tau^{*+} = 2$  against  $\sigma^+ = 1$ . And  $\tau^{*-}$  was missing  
 126 entirely.

127 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric  
 128 saturator that meets (4) with equality and attains its extremes there ( $\Lambda_t^\pm = A_t^\pm = \pm b$ ; the harness's  
 129  $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$  is one, Section 2), and to a constant  $\hat{q}$ . Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

130 with the endpoint admissible only when  $\hat{q} < 0$ : exactly Corollary 2's radius at  $\mu_t = 1$ , now as an  
 131 if-and-only-if. **The necessity direction above assumes none of it:**  $\tau > \tau^{*+}$ , or  $\tau \geq \tau^{*-}$ , forfeits  
 132 coverage against every admissible  $r_t$  and every legal predictable  $\{\hat{q}_t\}$ , extremes attained or not, and  
 133 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather  
 134 than decorative: the unattained supremum and the varying- $\hat{q}$  split two paragraphs above are exactly  
 135 what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they  
 136 always do.

137 **The mirror boundary is already measured, on this paper's own grid.** The  $\tau$ -cliff run's exact-  
 138 threshold column under the strict indicator is the mirror adversary  $s_t \equiv -b/2$  (one error process,  
 139  $\text{err}_t = \mathbf{1}\{\hat{q}_t < -b/2\}$ ), and on the same eleven widths at the null scorecaster it locates  $\tau^{*-} = 1$ ,  
 140 closed on the failing side, exactly where (2) puts it; Appendix A prints the eleven readings. **So the**  
 141 **failure criterion is  $|E_T|$ , not the miscoverage rate.** Both 1.000000 and 0.000000 are failures of the  
 142 same order,  $|E_T| = (1 - \alpha)T$  and  $\alpha T$ ; and once  $\hat{q}$  varies the rate stops being a tell at all.

143 **Both degenerate edges, and what they need.** An unbounded supremum is not on its own enough  
 144 for  $\tau^{*+} = \infty$  to mean what it looks like: the same unreachable-set construction with  $r_t(x) = x$   
 145 past  $|x| > t^2$  has no finite supremum and fails at  $\tau = 1.5$ . What the tangent family has, and that  
 146 construction does not, is a genuine reach:  $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$ , so the accumulator  
 147 level at which the deployed threshold is forced to  $b/2$  is  $\kappa(\tau)ch(t)$  with  $\kappa(\tau) = \arctan(b/2 +$   
 148  $\tau - \hat{q})/\arctan(b)$ , finite at every finite  $\tau$ . Coverage is retained at every width, as measured; but  
 149  $\kappa(\tau) > 1$  once  $\tau > b/2 + \hat{q}$ , so Proposition 2's own *constant* is forfeited past  $\tau = b/2$  while coverage  
 150 is not. At  $\tau = 1.5$ ,  $T = 10^6$  the excursion is 15.10 against Proposition 2's 14.8155 and against  
 151  $\kappa(1.5)ch(T) + 1 = 15.8530$ . At the other edge,  $\sup_t \hat{q}_t = -b/2$  forces  $\hat{q} \equiv -b/2$ , since every  $\hat{q}_t$   
 152 is at least  $-b/2$  already. There  $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$  for every  $\tau > 0$  and every  $t$ , and  
 153  $\hat{q}_1 = 0 < b/2$ , so the failing set is absorbing from round 1 with no transient: the admissible window  
 154 is the single point  $\tau = 0$ , and  $\tau = 0.5$  returns 1.000000, as does partial adjustment at  $w = 0.999$ .

155 **What that establishes is that the published condition is tight.** At  $\mu_t = 1$ ,  $\hat{q} \equiv 0$  the boundary  
 156 is exactly Corollary 2's radius, crossing it costs coverage rather than the rate ( $\tau = 0.9$  covers at  
 157 0.100012,  $\tau = 1.5$  returns 1.000000 with  $\max_t |E_t| = 900,000$ , 60,747 times Proposition 2's bound  
 158 at  $T = 10^6$ ; Table 2, App. A), and (2) makes that an if-and-only-if against the whole legal adversary  
 159 class rather than a pattern across five settings. **The claim is that necessity: a tightness result**  
 160 **about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by  
 161 post-processing. Elsewhere it stays sufficient only, and now provably so: under the tangent integrator  
 162 Corollary 2 permits  $\tau \leq b/2$  while no finite width forfeits coverage against any legal adversary,  
 163 because the reach is genuine at every level. At  $\hat{q} \equiv +b/2$  the looseness goes the other way and  
 164 disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary,  
 165 the unsmoothed control included, though  $\tau = 1.5$  covers there against the specified one. That  
 166 asymmetry makes "tight" a located claim.

167 **The constructive reading, and it is the corollary's rather than ours.** Placement B keeps  $\{\hat{q}_t\}$   
 168 inside the admissible radius, so Theorem 1 carries it already. What Section 3 adds is that the band  
 169 has two edges, both computable before deployment from the integrator's guaranteed reach and the  
 170 scorecaster's range, that crossing either is not a graceful degradation, and that between  $\sigma^\pm$  and  $\tau^{*\pm}$ ,  
 171 which separate only when  $\hat{q}$  varies in  $t$ , neither direction is settled here.

## 4 Limitations

**The retention direction is proved with no gap only where the two boundaries meet:** a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying  $\hat{q}$  leaves the band  $\sigma^+ < \tau \leq \tau^{*+}$  open in neither direction (Section 3). And  $0.63/(1-w)$  is specific to the saturator’s *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator  $h(t) = t/\log t$ , the certified gap  $(ch(T) + 1)/T$  decays as  $O(1/\log T)$  rather than  $O(1/T)$ . What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.**  $\tau$  is in score units with  $b = 2$ , so the failing  $\tau = 1.5$  is  $0.75b$ . Under i.i.d. scores it returns 0.249376 at  $T = 10^6$ , an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at  $b - \tau = 0.5$ , and uniform scores on  $[-b/2, b/2]$  put  $P(s > 0.5) = 0.25$  exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed** (20260820,  $T = 10^6$ ); **the primary, adversarial harness uses none, and only its null-scorecaster edge is bracketed;** the others are one-sided. **What remains open is the band between the two boundaries.** For a genuinely time-varying  $\hat{q}$ , coverage in  $\sigma^+ < \tau \leq \tau^{*+}$  is neither proved nor falsified; Table 3’s power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

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## 292 Code and data

293 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-  
 294 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])  
 295 [LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

## 296 A Supporting measurements

297 **The primary regime.** Every number in Table 2 is measured under one deterministic adversary,  
 298 which consumes no randomness and needs no seed. It plays  $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$  with  $\varepsilon = 10^{-9}$   
 299 against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and  
 300 the distinction is measured rather than argued. Scoring an exact-threshold play as a *miscoverage*  
 301 reproduces every quantity Table 2 reports: 44 of 44 arm-horizon cells identical, all eleven arms at  
 302 all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every  
 303 row, because the adversary’s signature move then covers rather than misses. Partial adjustment at  
 304  $w = 0.999$  excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place  
 305 of 1.000000. The  $\varepsilon \downarrow 0$  limit the harness specifies is the first reading, and that 623.70 turns on the  
 306 adversary’s  $\varepsilon$ , not the counting convention. **No row is convention-free with respect to the second,**  
 307 which is why the adversary is specified rather than tie-broken. At  $\tau = \tau^{*+}$  itself the two readings part:  
 308 the strict indicator returns 0.000000 where scoring the exact play as a miscoverage returns 1.000000.  
 309 **Neither of those is coverage.** They are  $|E_T| = \alpha T$  and  $(1 - \alpha)T$ , failures of the same order and  
 310 opposite sign. Under the strict indicator the exact-threshold play scores  $\text{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$ ,  
 311 which is also the indicator of the constant legal play  $s_t \equiv -b/2$ , so that column is Section 3’s *mirror*

Table 2: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to  $T = 10^6$ ; all eleven arms run are listed. A four-arm subset of this table – the control, the strongest partial-adjustment arm, and the two dead-band widths bracketing  $\tau^{*+}$  – appears in the body as Table 1; this table adds the remaining seven arms, the  $T = 2 \times 10^5$  horizon, saturation fraction, and threshold travel the body table omits. Proposition 2’s bound is 13.2061 at  $T = 2 \times 10^5$  and 14.8155 at  $T = 10^6$ ; “ratio” is  $\max_t |E_t|$  over it, and “Sat.” the fraction of rounds on which (4) delivers its full  $\pm b$ . Every number is the *primary* regime’s, defined in this appendix, which also gives the  $T = 10^4$  excursions and the fitted excursion law.

| Readout on the completed threshold | miscov. $2 \times 10^5$ | miscov. $10^6$  | $\max_t  E_t  \cdot 10^6$ | ratio  | Sat. $10^6$ | travel $10^6$ |
|------------------------------------|-------------------------|-----------------|---------------------------|--------|-------------|---------------|
| none (control)                     | 0.100035                | 0.100007        | 7.80                      | 0.53   | 0.0000      | 28,311        |
| partial adj. $w = 0.5$             | 0.100030                | 0.100007        | 7.70                      | 0.52   | 0.0000      | 18,113        |
| partial adj. $w = 0.9$             | 0.100030                | 0.100007        | 7.50                      | 0.51   | 0.0000      | 4,081         |
| partial adj. $w = 0.99$            | 0.100040                | 0.100006        | 63.00                     | 4.25   | 0.0007      | 1,023         |
| partial adj. $w = 0.999$           | 0.100035                | <b>0.100004</b> | 623.70                    | 42.10  | 0.0097      | 202           |
| growing time constant $p = 0.5$    | 0.100030                | 0.100006        | 12.60                     | 0.85   | 0.0000      | 330           |
| growing time constant $p = 0.9$    | 0.100030                | 0.100024        | 53.00                     | 3.58   | 0.3181      | 16            |
| running mean                       | 0.099940                | 0.100010        | 49.50                     | 3.34   | 0.3647      | 9             |
| dead band $\tau = 0.5$             | 0.100035                | 0.100005        | 10.60                     | 0.72   | 0.0000      | 2,045         |
| dead band $\tau = 0.9$             | 0.100060                | 0.100012        | 13.20                     | 0.89   | 0.0038      | 1,051         |
| dead band $\tau = 1.5$             | <b>1.000000</b>         | <b>1.000000</b> | 900,000                   | 60,747 | 1.0000      | 0.5           |

adversary rather than a second convention. It returns between 0.09992 and 0.09997 at  $\tau = 0.5, 0.9$ , 0.95, 0.99 and 0.999, and 0.000000 with  $\max_t |E_t| = 10,000$  at  $\tau = 1, 1.001, 1.01, 1.05, 1.1$  and 1.5. That locates  $\tau^{*-} = 1$  on the same eleven widths Figure 2 uses, closed on the failing side. The convention decides which legal adversary realises the endpoint failure, not whether one occurs.

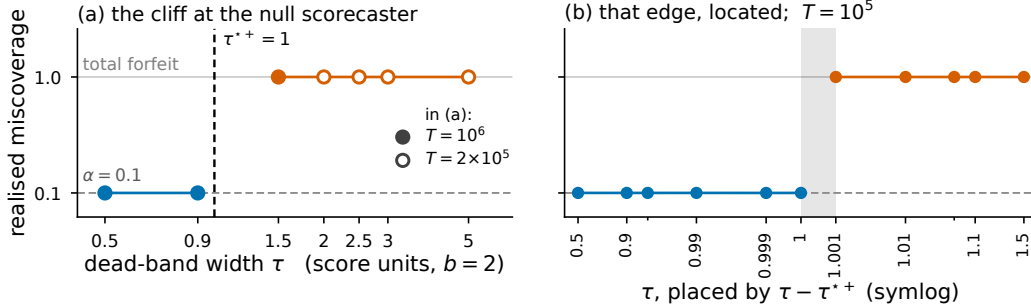


Figure 2: Realised miscoverage against dead-band width  $\tau$  at the null scorecaster,  $\hat{q} \equiv 0$  with a saturator of level  $b$ , under the *specified* adversary, where the upward boundary  $\tau^{*+}$  of (2) is 1. (a) The full sweep: filled markers run to  $T = 10^6$  and the four widest bands, drawn open, to  $T = 2 \times 10^5$ ; blue covers, orange forfeits, and the dashed rule is  $\tau^{*+}$ . (b) The same setting at  $T = 10^5$  over eleven measured widths, placed by  $\tau - \tau^{*+}$  on a symlog axis whose linear zone is  $|\tau - \tau^{*+}| \leq 10^{-3}$ , which magnifies the neighbourhood of  $\tau^{*+}$ : visual spacing here is not proportional to true distance in  $\tau$ . Unlabelled minor ticks mark  $\tau = 0.95$  and  $1.05$ . The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four  $(r_t, \hat{q})$  settings. The same eleven widths under the *mirror* adversary, which locates  $\tau^{*-}$ , are read off in this appendix’s primary-regime paragraph.

**Excursions at  $T = 10^4$ , and where the excursion law holds.** Partial adjustment excurses 6.30, 63.00 and 623.70 at  $w = 0.9, 0.99$  and  $0.999$ . The last two are flat in  $T$  while Proposition 2’s bound grows like  $\log T$ , which is why the ratio falls with the horizon: at  $w = 0.999$  it is 61.08 at  $T = 10^4$  and 42.10 at  $T = 10^6$ . On the control and the four partial-adjustment arms the excursion fits  $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$  at every horizon, most loosely at  $w = 0.9$ , the one arm whose two branches cross inside the horizon range. This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing time constants, the running mean and the dead bands all excurse further than it predicts.

**Realised miscoverage at the covering settings of Figure 3.** At  $\hat{q} \equiv +b/2$  the  $\tau = 1.5$  band returns 0.100010 with  $\max_t |E_t| = 11.20$ , inside the bound. Under Angelopoulos et al.’s tangent integrator the same width returns 0.100001. Raising the saturator’s level to  $4b$  leaves coverage intact and cuts

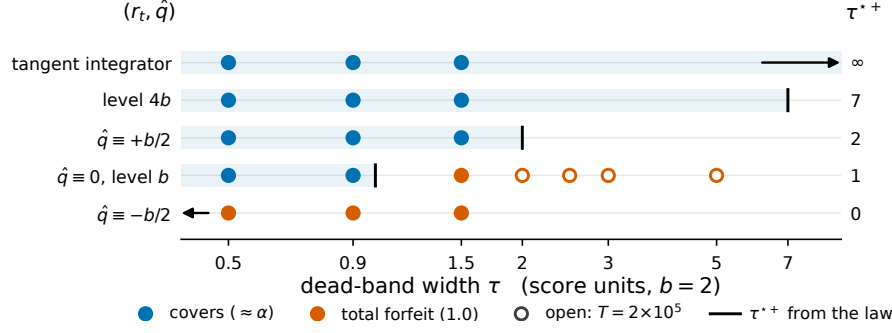


Figure 3: All 19 dead-band runs, at the five admissible  $(r_t, \hat{q})$  settings: rows are settings ordered by  $\tau^*$ , and the shared horizontal axis is  $\tau$ . Blue markers cover at  $\approx \alpha$  and orange markers return 1.000000; filled markers run to  $T = 10^6$  and open markers to  $T = 2 \times 10^5$ , all under the *specified* adversary. The rule on each row is that setting's upward boundary  $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ , printed at the right, and the tinted band is  $\tau \leq \tau^{*+}$ . Two rules fall off the axis and are drawn as arrows:  $\tau^{*+} = 0$  at  $\hat{q} \equiv -b/2$ , and  $\tau^{*+}$  unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary  $\tau^{*-}$  of (2) is not drawn here and differs from  $\tau^{*+}$  at the two non-null scorecasters.

the  $w = 0.999$  excursion from 623.70 to 120.60. That is the first term of  $\tau^*$  moving the edge, not the readout changing behaviour.

**What survives of the measurement, and what its status now is.** Outside the admissible window the transition is a step. Past  $\tau^{*+}$  the threshold sits permanently below the reachable score range and  $|E_t| = (1 - \alpha)t$  grows linearly, 60,747 times the bound at  $T = 10^6$ ; the wider bands  $\tau = 2.0, 2.5, 3.0$  and  $5.0$  fail too at  $T = 2 \times 10^5$  (Figure 2). **Here**  $\sup_x r_t(x) = b$  and  $\hat{q} \equiv 0$ , so  $\tau^{*+} = \tau^{*-} = b/2$ , one point of the admissible class, and both terms move it. The legal  $\hat{q} \equiv +b/2$  carries the  $\tau = 1.5$  band inside the bound against the specified adversary, and against the mirror one no width covers there, the unsmoothed control included. Condition (4) bounds  $\sup_x r_t(x)$  only from below, so raising it costs the failure direction nothing; it moves the edge too (Figure 3) and shrinks the excursion, from 623.70 to 20.10 under the tangent integrator Angelopoulos et al. [2023] report using in all their experiments, both at  $T = 10^6$ . **The method's own integrator therefore sits inside the set at every width run here. The endpoint is where the two boundaries part.** The upper one is closed and the lower one is open, which the null scorecaster cannot show because  $\tau^{*+}$  and  $\tau^{*-}$  coincide there: four saturators with  $\tau^{*+} < \tau^{*-}$  cover at  $\tau = \tau^{*+}$  exactly under both adversaries, and at  $\hat{q} \equiv +0.4$ , where  $\tau^{*-} = 0.6 < 1.4 = \tau^{*+}$ , the width 0.599 covers under both while 0.6 fails under the mirror. At the baseline the two coincide at 1, so the primary regime covers there and the mirror adversary does not; Figure 2 and this appendix's primary-regime paragraph print both readings. **The failing arms have** Sat. = 1.0000. Condition (4) delivers its full  $\pm b$  on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop is the integrator's output.

**Where the rate stops being a tell.** At  $\tau = 1.9$  with the power-of-two scorecaster the rate is 0.110960 against  $\alpha = 0.1$  while  $\max_t |E_t|$  is 84,745.70 at  $T = 10^6$ , so a reader watching only the rate would call that coverage.

Table 3: Legal  $(r_t, \hat{q})$  configurations on which the printed  $\tau^*$  over-certifies, each under two legal adversaries. “Proved window” is where (2) certifies retention against every legal adversary; between it and the failure boundaries  $\tau^{*\pm}$  neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here  $\emptyset$  marks a configuration for which no width is certified,  $\tau = 0$  included.  $T = 10^5$  except the two power-of-two rows, at  $T = 10^6$ . The last row carries no dead band at all.

| $(r_t, \hat{q})$ configuration        | $\tau$ | printed $\tau^*$ | proved window | $s_t = \tilde{q}_t + \varepsilon$ |                | $s_t \equiv -b/2$ |                |
|---------------------------------------|--------|------------------|---------------|-----------------------------------|----------------|-------------------|----------------|
|                                       |        |                  |               | miscov.                           | $\max_t  E_t $ | miscov.           | $\max_t  E_t $ |
| $\hat{q} \equiv +0.4$ , level $b$     | 0.599  | 1.4              | $\tau < 0.6$  | 0.100010                          | 7.20           | 0.099890          | 11.60          |
| $\hat{q} \equiv +0.4$ , level $b$     | 0.8    | 1.4              | $\tau < 0.6$  | 0.100050                          | 8.70           | <b>0.000000</b>   | 10,000         |
| $\sup_x r_t = 10b$ , off the reach    | 1.001  | 19               | $\tau < 1$    | <b>1.000000</b>                   | 90,000         | 0.000000          | 10,000         |
| $\sup_x r_t$ unbounded, off the reach | 1.5    | $\infty$         | $\tau < 1$    | <b>1.000000</b>                   | 90,000         | 0.000000          | 10,000         |
| $A^+ = 4b$ , $A^- = -b$               | 1.5    | 7                | $\tau < 1$    | 0.100020                          | 4.30           | <b>0.000000</b>   | 10,000         |
| level $4b$ , $\hat{q} \equiv 0$       | 6.999  | 7                | $\tau < 7$    | 0.100070                          | 12.20          | 0.100050          | 11.60          |
| level $4b$ , $\hat{q} \equiv 0$       | 7.0    | 7                | $\tau < 7$    | 0.100090                          | 12.20          | <b>0.000000</b>   | 10,000         |
| $\sup_x r_t = 2b$ , unattained        | 2.99   | 3                | $\tau < 1$    | 0.100320                          | 211.10         | 0.098010          | 210.50         |
| $\sup_x r_t = 2b$ , unattained        | 3.0    | 3                | $\tau < 1$    | <b>1.000000</b>                   | 90,000         | 0.000000          | 10,000         |
| $\sup_x r_t = 2b$ , attained          | 3.0    | 3                | $\tau < 3$    | 0.100000                          | 12.20          | <b>0.000000</b>   | 10,000         |
| $\hat{q}_t = b/2$ on powers of 2      | 1.0    | 2                | $\emptyset$   | 0.100009                          | 14.30          | <b>0.000000</b>   | 100,000        |
| $\hat{q}_t = b/2$ on powers of 2      | 1.9    | 2                | $\emptyset$   | <b>0.110960</b>                   | 84,745.70      | 0.000000          | 100,000        |
| $\hat{q} \equiv +b/2$ , no band       | 0      | 2                | $\emptyset$   | 0.100000                          | 0.90           | <b>0.000000</b>   | 10,000         |

## B The four vocabularies

Table 4: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness, convergence index, (in)consistency*; and *switching cost*. The row gives what each contributes.

| Online conformal                                                                                                               | Forecast stability                                                                                                                                   | Verification                                                                                | Switching-cost online learning                                           |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Distribution-free long-run coverage for any bounded scorecaster as post-processing [Angelopoulos et al., 2023, Thm 1, Prop. 2] | Movement functional; readout as post-processing [Godaheva et al., 2025, Eq. 3]; priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026] | Matched-level design, real producers (2012) [Pinson and Girard, 2012, Worsnop et al., 2018] | Worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2] |

## C Where this sits

Four vocabularies name one quantity, and Table 4 names all four; this appendix places this paper’s claim and its concessions among them. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that  $\sum_{t \leq T} q_t \leq O(\alpha h(T))$  for a sublinear  $h$ , supported with a plot rather than derived. Verification’s matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing  $\theta$  with  $T$ , which grows the ratio [Andrew et al., 2013, Thm. 7].

**The accumulator, read as a bet.** Appendix D reads  $E_t$  as a game-theoretic capital process and builds a calibration test on top of the boundary above; it is not repeated here.

## D The accumulator as a bet, and a calibration test built on the boundary

**What is assumed here, and by whom.** Everything Section 3 proves is deterministic and pathwise. Proposition 2’s window  $|E_T| \leq c h(T) + 1$ , and the boundaries (2) that locate its edge, hold on every trajectory a legal adapted adversary can produce; no distributional model on the data appears in either statement, and neither is a statement about a null hypothesis. This section neither uses that nor weakens it. It runs the other way: it takes the error sequence a deployment actually produced as *data*, and tests a hypothesis about it. The hypothesis is imposed here, for the test alone, with  $\mathcal{F}_{t-1}$  the history through round  $t - 1$  (scores, deployed thresholds and errors) against which  $\hat{q}_t$  is already required to be predictable:

$$H_0 : \quad \text{err}_t \mid \mathcal{F}_{t-1} \sim \text{Bernoulli}(\alpha) \quad \text{for every } t,$$

and it says that the shipped intervals were conditionally calibrated at the target rate.  $H_0$  is the object under test. It is not a premise of Proposition 2, of (2), or of any other claim in this paper; nothing above requires it, implies it, or is disturbed if it fails. When it fails, what loses validity is the test below and nothing else. We are explicit about this because the rest of the paper assumes no distribution at all, and a reader who meets a Bernoulli null on this page must not read it backwards into the pages before it.

**The accumulator is a capital process, and it is signed.** A Skeptic who each round pays  $\alpha$  for the payoff  $\text{err}_t$  has cumulative net gain exactly  $E_t$ , so this paper’s accumulator is a game-theoretic capital

process at unit stake [Shafer and Vovk, 2019, Ramdas et al., 2023]. It is not an e-value and not an e-process: it is signed, it goes negative on exactly the trajectories the mirror boundary  $\tau^{*-}$  governs, and Proposition 2 bounds it pathwise by feedback rather than under a null by Ville’s inequality [Vovk and Wang, 2021]. Nothing below proposes  $E_t$ , or a transform of it, as a certificate for any claim this paper makes. What is formal is the standard transform, and it is invoked rather than invented.

**The test martingale, and why it is one.** Fix  $\lambda$  and set  $M_0 = 1$ ,

$$M_t = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha)).$$

Each factor takes one of two values,  $1 + \lambda(1 - \alpha)$  on a miss and  $1 - \lambda\alpha$  on a cover, so  $M_t \geq 0$  for every path if and only if  $\lambda \in [-1/(1 - \alpha), 1/\alpha]$ , which for  $\alpha = 0.1$  is  $[-10/9, 10]$ . Under  $H_0$ ,  $\mathbb{E}[\text{err}_t | \mathcal{F}_{t-1}] = \alpha$ , so each factor is conditionally mean one and  $\mathbb{E}[M_t | \mathcal{F}_{t-1}] = M_{t-1}$ :  $M_t$  is a nonnegative martingale started at 1, i.e. a test martingale for  $H_0$ . Ville’s inequality then gives  $\mathbb{P}_{H_0}(\sup_t M_t \geq 1/\eta) \leq \eta$ , so “reject  $H_0$  the first time  $M_t \geq 1/\eta$ ” is an anytime-valid level- $\eta$  test of the deployed intervals’ calibration, at any stopping time and with no horizon fixed in advance. That is a test of the deployment, not of Section 3.

**The boundary transfers exactly, because  $M_t$  sees only  $(t, E_t)$ .** Since  $\text{err}_i \in \{0, 1\}$ , the product collapses. Writing  $N_t$  for the miss count and using  $N_t = \alpha t + E_t$ ,

$$\log M_t = K(\lambda)t + L(\lambda)E_t, \quad K(\lambda) = \alpha \log(1 + \lambda(1 - \alpha)) + (1 - \alpha) \log(1 - \lambda\alpha), \quad L(\lambda) = \log \frac{1 + \lambda(1 - \alpha)}{1 - \lambda\alpha},$$

with  $K(\lambda) \leq 0$ ,  $K(\lambda) = 0$  only at  $\lambda = 0$ , and  $\text{sign } L(\lambda) = \text{sign } \lambda$ . So  $M_t$  depends on the path only through the horizon and the accumulator – the same statistic Section 3’s dichotomy is stated in – and the transfer is an identity rather than the first-order expansion  $\log M_t = \lambda E_t + O(\lambda^2 t)$ . It transfers as a dichotomy, not as a measured rate. Inside the retained band  $|E_t| \leq c h(t) + 1$  is logarithmic, so  $\log M_t \leq K(\lambda)t + |L(\lambda)|(c h(t) + 1)$  at every round. Since  $K(\lambda) < 0$  the right side falls to  $-\infty$  linearly, so  $M_T \rightarrow 0$  exponentially; and since it is continuous with a single interior maximum it has a finite supremum  $S(\lambda)$  over all  $t$ , so  $\sup_t M_t \leq e^{S(\lambda)}$  uniformly in the horizon. The test therefore never rejects at any level  $\eta \leq e^{-S(\lambda)}$ , however long the deployment runs. It is the uniform bound and not the limit that non-rejection needs, because Ville’s inequality prices the supremum. Past either boundary  $|E_T|$  is linear in  $T$ , so some fixed  $\lambda$  of the sign of  $E_T$  makes  $\log M_T$  grow linearly and the test rejects at any fixed level within  $O(1)$  rounds. Both signs are needed, and that is the point of the two-sided form: the  $\tau^{*+}$  failure drives  $E_T$  up and calls for  $\lambda > 0$ , the mirror failure at  $\tau^{*-}$  drives it down and calls for  $\lambda < 0$ .

**Worked on this paper’s own frozen arms, at the null scorecaster.** Take the harness of Section 2 ( $\alpha = 0.1$ ,  $b = 2$ ,  $c = 1$ ,  $h(t) = \log(t + 2)$ ,  $\hat{q} \equiv 0$ ), where  $\tau^{*+} = \tau^{*-} = 1$ , and the two dead bands that bracket that edge in Table 2 (App. A). *Past the edge*,  $\tau = 1.5$  returns miscoverage 1.000000 and  $\max_t |E_t| = 900,000$  at  $T = 10^6$ ; the run records  $n_{\text{err}} = T$  at all four horizons, so the per-round path is exactly  $\text{err}_t = 1$  and  $E_t = (1 - \alpha)t$ , and no reconstruction is involved. At  $\lambda = 1$  – a round number well inside  $[-10/9, 10]$ , not the aggressive endpoint – the identity above gives  $M_t = 1.9^t$ , so at level  $\eta = 0.05$ , threshold  $1/\eta = 20$ :  $M_4 = 13.03$  and  $M_5 = 24.76$ . **The anytime-valid test rejects on the fifth deployed interval.** At  $\eta = 0.001$  it rejects on the eleventh ( $M_{10} = 613.11$ ,  $M_{11} = 1164.90$ ). *Inside the edge*,  $\tau = 0.9$  covers at 0.100012 with  $\max_t |E_t| = 13.20$  against Proposition 2’s 14.8155 at the same  $T = 10^6$ , and the same  $\lambda = 1$  gives  $\log M_t \leq -0.030639t + 9.8632$  from the measured excursion alone: below  $\log 20$  from round 225, below 0 from round 322, and  $M_T \leq e^{-30,629}$  at  $T = 10^6$ . The measured excursion bounds one run; the theorem bounds every one. This width sits inside  $\sigma^\pm$ , so the retention half applies and  $|E_t| \leq c h(t) + 1$  holds at every round, whence  $S(1) = \sup_t (K(1)t + L(1)(\log(t + 2) + 1)) = 2.448$ , attained at  $t = 22$ , and  $M_t \leq 11.57$  at every round and every horizon. The test cannot reject at  $\eta = 0.05$  – not merely after round 225, but ever. Five rounds against never, on two arms of one frozen grid. *The mirror direction*, on the same  $\tau = 1.5$  configuration read under the strict indicator against  $s_t \equiv -b/2$ : miscoverage 0.000000,  $\text{err}_t = 0$  at every round,  $E_t = -\alpha t$ , and  $\lambda = -1$  gives  $M_t = 1.1^t$ , which crosses 20 at  $t = 32$  ( $M_{31} = 19.19$ ). The asymmetry is not a defect of the test: a forced miss carries  $\log 1.9$  nats per round against a forced cover’s  $\log 1.1$ , because  $\alpha = 0.1$  makes misses the rarer event under  $H_0$ .

437 **What this adds, and what it does not.** It adds a diagnostic and a reading, not a result. The  
438 reading is that  $E_t$  is the Skeptic’s capital and that Section 3 therefore locates, in a quantity a bettor  
439 already cares about, the exact width at which a dead band turns a calibrated deployment into one that  
440 an anytime-valid test convicts within a handful of rounds. The diagnostic is runnable online by a  
441 practitioner who has the errors and not the proof. Neither statement is a claim about  $E_t$ ’s status:  $E_t$  is  
442 not an e-process,  $M_t$  is a textbook transform imposed on top of the result rather than a part of it, and  
443 this paper’s contribution is the tightness of someone else’s admissible radius, proved pathwise.  $M_t$  is  
444 itself an e-value, and we say so: an e-value for  $H_0$  alone, a hypothesis about one deployment’s error  
445 sequence, imposed on this page and nowhere else in the paper. No result of this paper is an e-value  
446 result or rests on one. Betting enters online conformal inference elsewhere as an optimiser rather  
447 than as evidence [Podkopaev et al., 2024].